

Xiaoshan Wu

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EDUCATION

Harvard University

Visiting Scholar (Embodied AI & 4D World Models)

Cambridge, MA, USA

Apr. 2026 – Present

– **Advisors:** Yilun Du and Mengyu Wang.

The University of Hong Kong (HKU)

Ph.D. in Electrical and Computer Engineering (Focus: 3D/4D Vision, World Models & Embodied AI)

Hong Kong

Sep. 2023 – Aug. 2027

– **Awards:** Hong Kong PhD Fellowship (HKPFS, **Top Tier**), HKU Presidential Scholarship.

– **Advisor:** Xiaojuan Qi. **GPA:** 4.0/4.3.

Zhejiang University (ZJU) & UIUC

Dual B.S. in Computer Engineering (ZJUI Institute)

China & USA

Sep. 2019 – Jun. 2023

– **Honors:** Outstanding Graduate (Rank 5), Dean's List. **GPA:** 3.99/4.0 (ZJU), 3.75/4.0 (UIUC).

RESEARCH INTERESTS

My research connects **3D/4D perception**, **generative world modeling**, and **memory for embodied intelligence**. My work spans geometry-grounded robot control, autoregressive video generation, and multi-view spatial reasoning. I am now interested in self-improving physical agents through agent-world model co-evolution.

PUBLICATIONS & PREPRINTS

* *Equal contribution; † project lead. Published, preprint, and under-review work.*

First- and Co-first-author Work

1. Verifiable 4D World–Action Model for Robust Robot Control.

X. Wu*, R. Cai*, Y. Bao, Z. Su, A. Wu, F. Wu, Z. Li, Z. Qi, H. Chen, W. Ye, J. Zhang, K. Zhou, M. Wang, and Y. Du.

Under review at ICLR 2027.

First author

S4NDBOX bridges visual imagination and robot control through explicit 4D prediction, guiding action generation and candidate selection by geometric agreement before execution.

RoboTwin Hard: +13.3 pp success over RGB-only after clean-only training; zero-shot sim-to-real on Stack Cups.

2. GaMe: Geometry as Memory for Multi-View Spatial Reasoning.

X. Wu, X. Lyu, Y. Yu, M. Liu, S. Shi, Z. Wang, and X. Qi.

Under review at ICRA 2027.

First author

3. VideoSSM: Autoregressive Long Video Generation with Hybrid State-Space Memory.

Y. Yu*, X. Wu*, X. Hu, et al.

arXiv:2512.04519, 2025.

Co-first author

4. EAG3R: Event-Augmented 3D Geometry Estimation for Dynamic and Extreme-Lighting Scenes.

X. Wu, Y. Yu, X. Lyu, et al.

NeurIPS 2025 • **Spotlight**.

First author

5. LiFR-Seg: Anytime High-Frame-Rate Segmentation via Event-Guided Propagation.

X. Wu, X. Lyu, Y. Yu, B. Wang, Z. Wang, and X. Qi.

ICLR 2026.

First author

6. EVSplat: Uncertainty-Aware Gaussian Splatting for Dynamic Scenes with Event Camera.

Y. Yu*, X. Wu*†, X. Lyu, Z. Wang, and X. Qi.

Under review at AAAI 2027.

Co-first author; project lead

7. LiFR v2: Completion-Augmented Event Propagation for High-Rate Dense Prediction.

T. Wan*, X. Wu*†, Y. Yu, et al.

Under review at IEEE TPAMI; arXiv:2609.25803, 2026.

Co-first author; project lead

8. Event-Based Depth Prediction With Deep Spiking Neural Network.

X. Wu, W. He, M. Yao, et al.

IEEE TCDS, 16(6):2008–2018, 2024.

First author

Collaborative Work

9. Efficient and Accurate Neural-Field Reconstruction Using Resistive Memory.

Y. Yu, X. Zhang, S. Wang, ..., X. Wu, et al.

Nature, 654:642–651, 2026.

10. FoundationGeo: Learning Spatial Pixel-Wise Fields for Monocular Metric Geometry.

M. Liu, X. Lyu, T. Ren, P. Dai, X. Wu, et al.

ECCV 2026.

11. Stabilizing Streaming Video Geometry via Dynamic Feature Normalization.

X. Lyu, M. Liu, X. Wu, et al.

CVPR 2026, pp. 7577–7587.

12. OptiGeo: Efficient Monocular Geometry for Embodied Perception in Optically Challenging Scenes.

M. Liu, T. Liu, J. Xia, X. Lyu, X. Wu, et al.

CoRL 2026.

13. Continuous-Time Digital Twin with Analog Memristive Neural Ordinary Differential Equation Solver.

H. Chen, J. Yang, J. Chen, ..., X. Wu, et al.

Science Advances, 11(22):eadr7571, 2025.

14. Topology Optimization of Random Memristors for Input-Aware Dynamic SNN.

B. Wang, X. Zhang, S. Wang, ..., X. Wu, et al.

Science Advances, 11(16):eads5340, 2025.

15. Charge-Selective 2D Heterointerface-Driven Multifunctional Floating Gate Memory for In Situ Sensing-Memory-Computing.

C. Li, X. Chen, Z. Zhang, X. Wu, et al.

Nano Letters, 24(47):15025–15034, 2024.

TECHNICAL SKILLS

PyTorch, CUDA, Diffusers, Open3D, OpenCV.

SELECTED AWARDS

Hong Kong PhD Fellowship (HKPFS, 2023); HKU Presidential Scholarship (2023).

Outstanding Graduate of Zhejiang University (2023); NeurIPS (2025) and ICLR (2026) Travel Awards.